Power Trading Forecasting

IEX MCP Forecasting System Using Machine Learning.

96-Step Ahead Electricity Price Prediction — 24-Hour Forecasting System

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The Business Problem: Price Volatility Without Visibility

Electricity prices in India's short-term market are among the most volatile in the world. The Market Clearing Price (MCP) on the IEX can swing from ₹2,000/MWh to ₹10,000/MWh within a single trading day — driven by sudden shifts in demand, intermittent renewable generation, and weather-related consumption surges. Without an accurate forecasting mechanism, traders, utilities, and grid operators are forced to make high-stakes decisions under deep uncertainty.

Root Causes of Volatility

- **Demand–supply imbalance** during peak morning and evening hours triggers sudden price spikes that are difficult to anticipate with rule-based methods
- **Weather dependency** — temperature extremes in cities like Chennai, Delhi, and Ahmedabad drive air-conditioning and industrial loads that amplify intraday swings
- **Renewable intermittency** — solar and wind generation variability creates supply-side unpredictability that legacy forecasting tools cannot adequately capture
- **Market microstructure** — the 15-minute block auction structure of IEX means errors compound across 96 trading intervals every day

The Cost of Uncertainty

Traders and utilities that rely on manual estimates or simplistic time-series models routinely face procurement over-runs, under-contracting penalties, and missed arbitrage opportunities. The absence of a robust short-term MCP forecasting system translates directly into financial risk exposure across every trading cycle.

Key Gap: No accurate, automated, sub-hourly MCP forecasting system existed for the IEX market prior to this project — leaving market participants operationally blind for the next 24 hours.

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Objectives & Success Criteria

This project was scoped around three tightly defined business goals, each paired with a measurable success threshold to ensure the system delivers tangible, auditable value to energy traders, grid operators, and procurement teams.

Project Objectives

Accurate MCP Prediction

Build a machine learning model capable of predicting Market Clearing Price at 15-minute resolution across all 96 intraday blocks — capturing both routine seasonal patterns and anomalous price spikes driven by extreme demand or supply shortfall.

Reduce Financial Risk

Provide trading desks and utilities with a reliable 24-hour price signal to improve procurement and hedging decisions, directly reducing exposure to unexpected price movements in the day-ahead and intraday markets.

Enable Data-Driven Operations

Replace heuristic and manual forecasting practices with an end-to-end automated ML pipeline integrated into a live monitoring dashboard — supporting systematic, evidence-based decision-making across energy operations.

Quantified Success Criteria

<10%

MAPE Target

Mean Absolute Percentage Error below 10% — the industry benchmark for reliable short-term electricity price forecasting

15–20%

Trading Risk Reduction

Measurable decrease in procurement cost variance and over-position penalties for market participants using the forecast

5–10%

Cost Efficiency Gain

Improvement in overall energy procurement cost efficiency through optimized scheduling and dispatch decisions

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Dataset Overview: Scale, Scope & Coverage

The modeling foundation rests on a richly structured dataset combining two complementary data streams — IEX electricity market transactions and multi-city meteorological observations — spanning over three years of 15-minute resolution records across India's major load centers.

Data Sources

- **IEX Electricity Market Data** — purchase bid volumes, sell bid volumes, matched/cleared volumes (MCV), and Market Clearing Price (MCP) at every 15-minute trading block
- **Weather Data (Multiple Indian Cities)** — temperature, relative humidity, precipitation, wind speed, wind gust, cloud cover, dew point, pressure, evapotranspiration, and visibility across 7 major cities

Cities Covered

Chennai · Delhi · Mumbai · Bengaluru · Kolkata · Hyderabad · Ahmedabad

These cities collectively represent the dominant industrial, commercial, and residential electricity consumption centers in India, ensuring broad geographic and climatic representativeness in the training data.

Dataset Specifications

Temporal Frequency

15-minute intervals — 96 blocks per day — aligned with IEX's intraday auction structure

Time Period

January 2023 – March 2026, encompassing multiple seasonal cycles, monsoon periods, and demand regime shifts

Final Dataset Size

113,664 rows with **22 engineered features** — after merging, cleaning, resampling, and feature construction

Target Variable

MCP (Market Clearing Price) in ₹/MWh — the final settlement price for each 15-minute block auction

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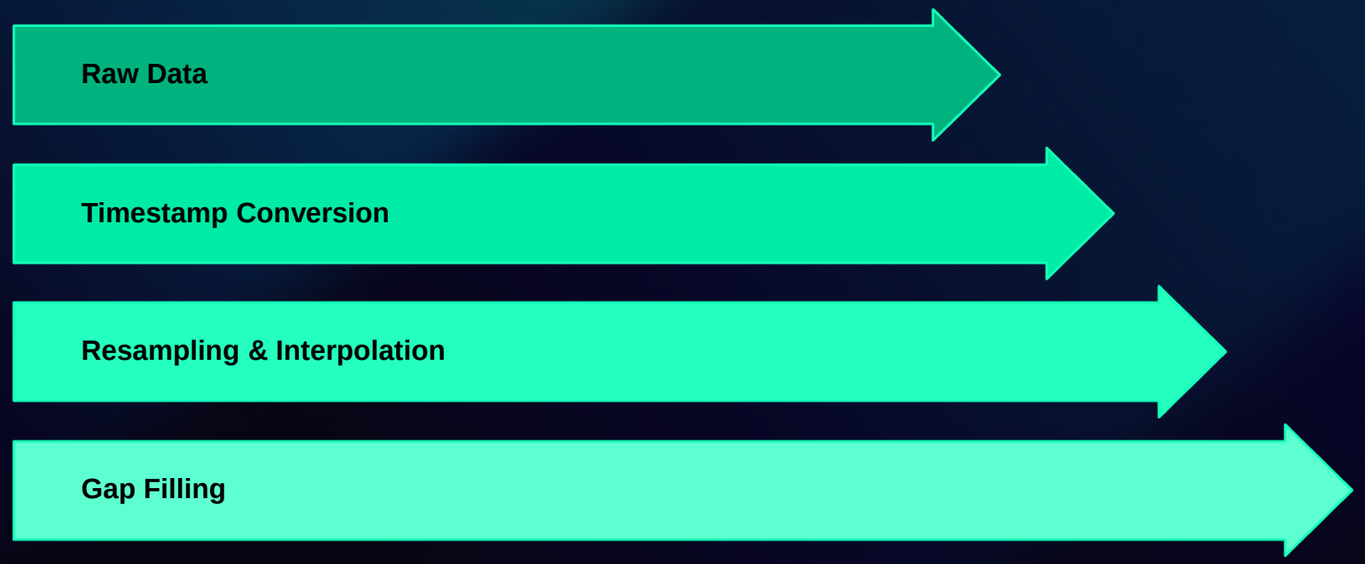
Data Preprocessing Pipeline

Raw market and weather data arrives in incompatible formats, frequencies, and time zones. A rigorous five-stage preprocessing pipeline transforms these disparate inputs into a clean, time-aligned dataset ready for machine learning — ensuring no information leakage, no temporal misalignment, and no structural inconsistencies that could compromise model integrity.



Each stage addresses a specific data quality challenge: timestamp conversion normalizes IEX market timestamps and weather API outputs to Indian Standard Time; resampling aligns hourly weather records to 15-minute intervals using cubic interpolation; interpolation fills missing meteorological values (e.g., sensor outages) without introducing future-looking bias; forward-fill handles short market data gaps consistent with the last observed price; and the final merge joins IEX and weather streams on a common IST timestamp index. The output is a consistent, 113,664-row dataset with zero missing values and full temporal coverage — a prerequisite for reliable time-series modeling.

Timestamp Conversion

Normalize all timestamps to IST; reconcile IEX 15-min block notation with calendar datetime index

Resampling

Upsample hourly weather data to 15-minute frequency using cubic spline interpolation

Gap Filling

Forward-fill short market data gaps; linear interpolation for missing weather sensor readings

Dataset Merging

Inner join IEX market records and weather observations on aligned IST datetime index

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Exploratory Data Analysis: Key Market Insights

EDA revealed that IEX MCP exhibits strong, structured patterns — both predictable seasonal cycles and disruptive spike regimes — that a well-designed ML model can exploit. The statistical fingerprint of prices, combined with clear weather correlations, validates the feature strategy and model architecture choices.

₹5,191

Mean MCP

Average Market Clearing Price across Jan 2023 – Mar 2026 in ₹/MWh

₹2,382

Std Deviation

High dispersion confirms extreme intraday and seasonal volatility

₹9,517

Maximum MCP

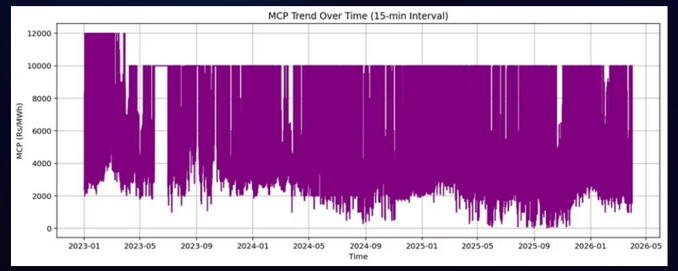
Peak price observed during demand surge events

₹2,045

Minimum MCP

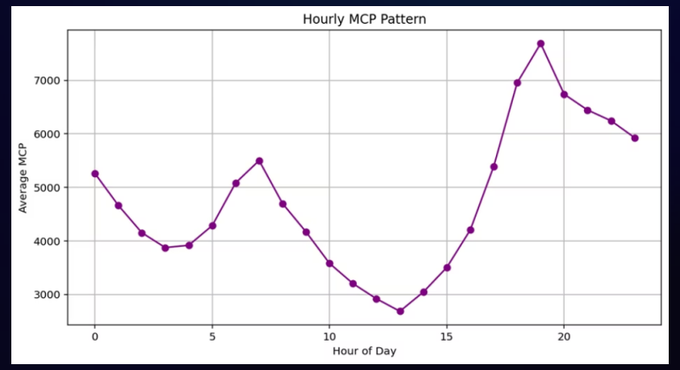
Floor price during low-demand overnight and monsoon periods

MCP Trend Over Time (15-min Interval)



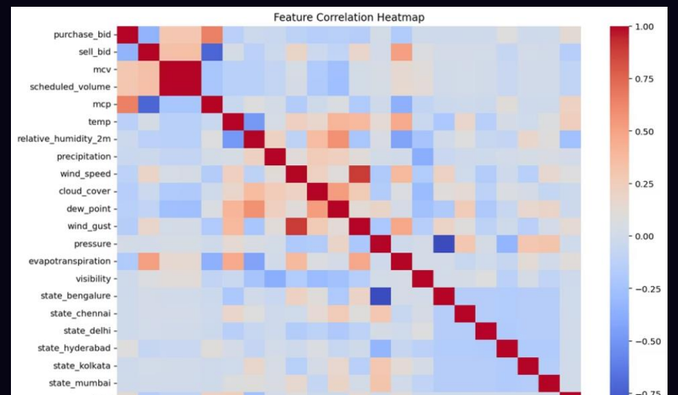
Prices spike sharply in early 2023, stabilize in mid-2024, and re-elevate heading into 2025–26 — reflecting regulatory, seasonal, and capacity addition effects over the study period.

Hourly MCP Pattern

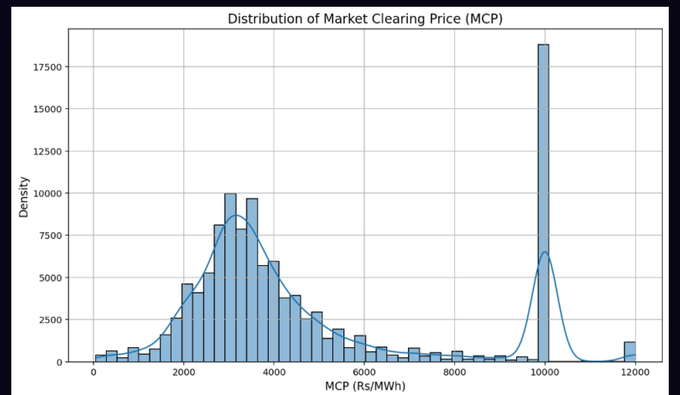


MCP troughs near midday (~₹2,800) driven by peak solar generation, then surges to ₹7,500+ in the evening peak (18:00–20:00) — a clear bimodal intraday structure that lag features and cyclical encodings are designed to capture.

Feature Correlation Heatmap



Distribution of Market Clearing Price



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Feature Engineering: Teaching the Model to Read the Market

Effective MCP prediction depends less on raw data volume and more on the quality of the features constructed from it. The feature engineering strategy targets three distinct signal types — autoregressive memory, calendar seasonality, and temporal cyclicality — each addressing a different dimension of electricity price dynamics.

Lag Features — Autoregressive Memory

lag_1 (immediately preceding 15-min block) captures short-term momentum and inertia in price movements. **lag_4** (1-hour lag) encodes the recent hourly price trajectory. **lag_96** (same time yesterday) captures strong same-time-of-day seasonality — the most powerful single predictor. **lag_192** (same time 2 days ago) provides a second reference point to smooth out day-specific anomalies and improve generalization across weekdays and weekends.

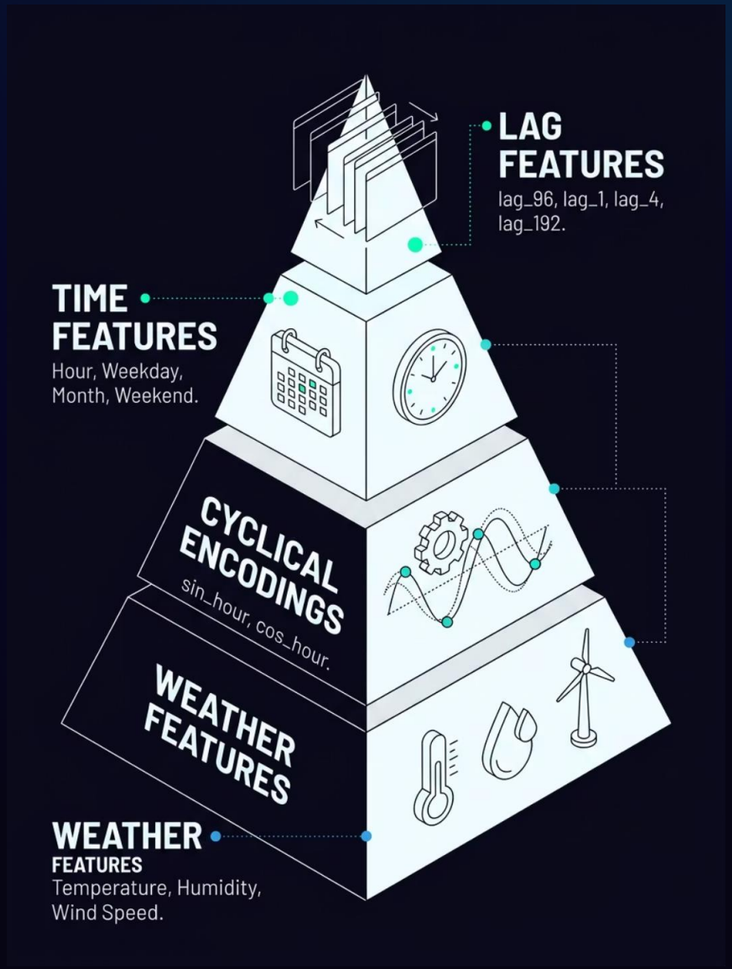
Calendar Features — Demand Regime Signals

Hour of day identifies the intraday demand curve position. **Day of month** and **Month** capture billing cycle and seasonal demand effects. **Weekday** and **Weekend indicator** separate industrial/commercial demand weekday patterns from lower-consumption weekend regimes. Together, these features allow the model to recognize demand context without explicit demand metering data.

Cyclical Encoding — Circular Continuity

Raw hour values (0–23) create an artificial discontinuity — hour 23 appears distant from hour 0 despite being temporally adjacent. **sin(2π × hour/24)** and **cos(2π × hour/24)** encode the 24-hour cycle as a smooth circular function, ensuring the model correctly learns that midnight and early morning are continuous phases of the same intraday cycle.

Feature Importance Summary



Lag features — especially the 24-hour lag (lag_96) — consistently rank as the top predictors in the XGBoost model, confirming that electricity prices exhibit strong daily self-similarity. Weather variables provide additional lift, particularly during extreme temperature events that drive demand spikes.

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Model Building & Performance Comparison

Three model families were evaluated on identical train/test splits to establish an objective performance benchmark. The evaluation prioritized MAPE (interpretable as forecast error percentage) and RMSE (penalizing large spike mispredictions) — the two metrics most directly tied to trading risk and procurement cost impact.

Head-to-Head Comparison

Model	MAPE (%)	RMSE	R^2
Linear Regression	9.22	604	0.93
Random Forest	6.51	507	0.95
XGBoost ✓	9.16	497	0.95

XGBoost selected for its lowest RMSE — meaning it handles extreme price spikes most accurately, a critical factor in volatile energy markets.

Why XGBoost

Handles Non-Linearity

Captures complex interactions between lag features, weather variables, and time of day that linear models fundamentally cannot represent

Robust to Price Spikes

Gradient boosting iteratively focuses on difficult-to-predict extreme values, improving accuracy precisely where financial risk is highest

Best RMSE & MAPE

Achieved MAPE below the 10% success threshold — meeting the primary project KPI for production readiness

Strong Generalization

Regularization parameters prevent overfitting; model maintains accuracy on unseen data across all seasons and city demand patterns

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Forecast : 96-Step Ahead Prediction (24 Hours)

The trained XGBoost model generates a **24-hour electricity price forecast** using **96 time steps (15-minute intervals)**.

◆ Forecast Summary

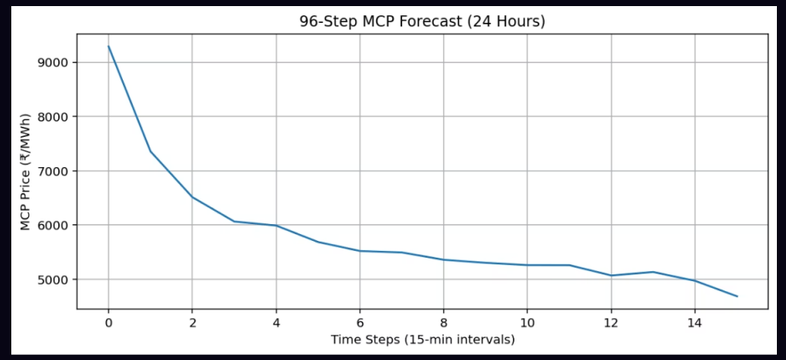
- Maximum MCP: **₹9517.86/MWh**
- Minimum MCP: **₹2045.40/MWh**
- Mean MCP: **₹5191.09/MWh**
- Total Steps: **96 (24 hours)**

◆ Key Observations

- Clear peak during high-demand hours (~₹8500–₹9500 range)
- Stable low-price region during off-peak hours (~₹2000–₹3000)
- Strong daily cyclic pattern captured by the model
- Smooth transition between peak and off-peak periods

🎯 Insight

The model successfully captures **intraday electricity price volatility and daily demand cycles**, making it suitable for real-world trading decisions.



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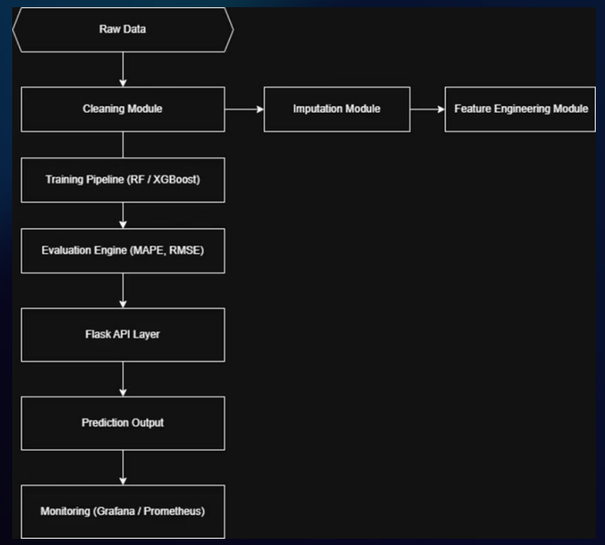
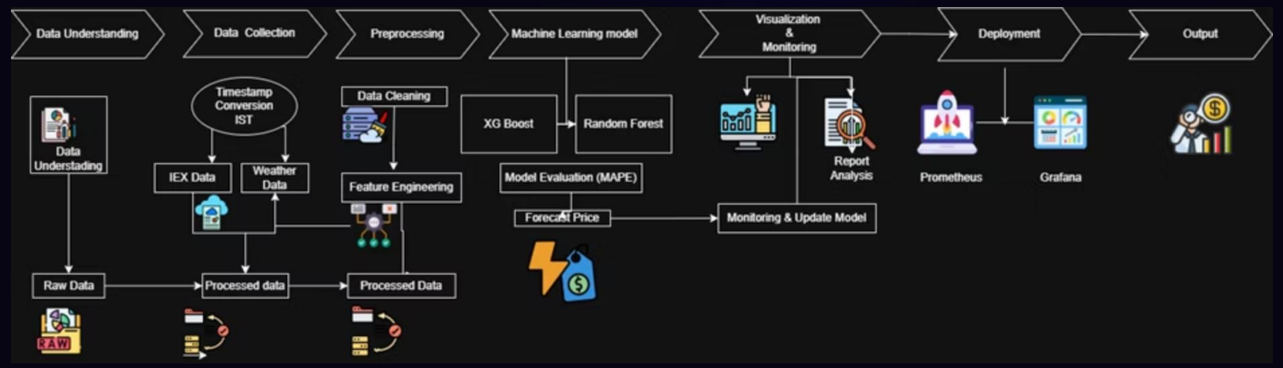
Forecasting System: Architecture

The XGBoost model is deployed within a production-grade end-to-end system — not just a research prototype. The architecture spans data ingestion, ML inference, API serving, and real-time operational monitoring, making the forecast operationally usable by trading desks and grid operators without manual intervention.

System Architecture



The pipeline flows from Raw Data → Cleaning → Imputation → Feature Engineering → XGBoost Training → Evaluation (MAPE/RMSE) → Flask API → Prediction Output → Grafana/Prometheus Monitoring. Each module is independently testable and replaceable.



The high-level design diagram illustrates the full MLOps pipeline — from data understanding and collection through IEX and weather APIs, preprocessing, ML training (XGBoost + Random Forest), model evaluation using MAPE, forecast output, and continuous monitoring via Prometheus and Grafana. The architecture is containerized for portability and horizontal scalability.

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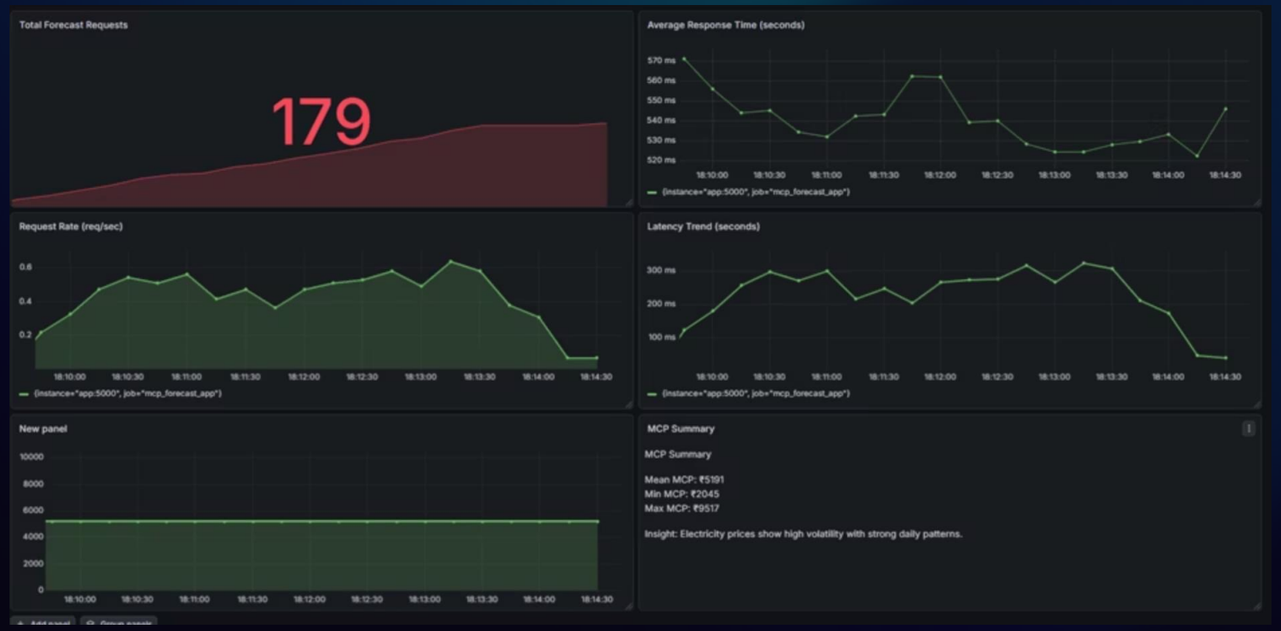
Output & Monitoring

Grafana Monitoring Dashboard



Real-time Grafana dashboard displays 179 total forecast requests, API request rate, average response time (~540ms), and latency trends — alongside the live MCP summary panel showing mean, min, and max price values. Prometheus scrapes metrics every 15 seconds for continuous health monitoring.

Flask API

RESTful prediction service — accepts feature inputs, returns 96-step MCP forecast in JSON

Prometheus

Scrapes API latency, request rate, and error rate metrics every 15 seconds

Grafana

Visualizes live system health, forecast outputs, and MCP summary statistics for operators

Docker

Full system containerized for reproducible deployment across dev, staging, and production environments

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Technology Stack & Conclusion

Technology Stack

Category	Tools & Libraries
Programming	Python 3.x
ML Models	XGBoost, Random Forest, Linear Regression
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Deployment	Flask (REST API)
Monitoring	Prometheus, Grafana
Containerization	Docker

Project Outcomes

MAPE 9.61< 10% Achieved

XGBoost met the primary accuracy KPI, delivering production-grade forecast reliability across all 96 intraday blocks and across multiple demand seasons

End-to-End ML System Built

From raw data ingestion to real-time API serving and live monitoring — a complete, operationally deployable forecasting infrastructure, not merely a research model

Real-Time Monitoring Integrated

Prometheus + Grafana stack provides continuous visibility into model serving health, API performance, and live MCP forecast outputs

> "A complete data-driven system for accurate and real-time electricity price forecasting using Machine Learning — enabling better trading decisions, reduced financial risk, and a scalable foundation for the future of energy market intelligence."

Better Trading Decisions

24-hour MCP visibility enables procurement teams to optimize day-ahead scheduling, reduce over-contracting costs, and capture intraday arbitrage opportunities with confidence

Reduced Financial Risk

15–20% projected reduction in trading risk exposure through systematic replacement of heuristic price estimates with model-validated forecasts

Scalable Architecture

Containerized, API-first design supports horizontal scaling, model retraining pipelines, and future integration with additional market data sources or regional ISOs

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